GENERAL ASSEMBLY

INTRODUCTION TO PROBABILITY

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DSI+

DATA SCIENCE PROCESS

1. Define problem.
2. Gather data.
3. Explore data.
4. Model with data.
5. Evaluate model.
6. Answer problem.

DEFINITIONS

- Experiment: A procedure that can be repeated infinitely many times and has a well-defined set of outcomes.

- Event: Any collection of outcomes of an experiment.

- Sample Space: The set of all possible outcomes of an experiment, denoted $\mathcal{S}$.

EXAMPLES

- Experiment: Flip a coin twice.
- Sample Space $\mathcal{S}$:
- Event:

- Experiment: Rolling a single die.
- Sample Space $\mathcal{S}$:
- Event:

DEFINITIONS

- Set: A well-defined collection of distinct objects.
  - $\{Derek\ Jeter, \pi, ☺\}$
    - (Standing on the shoulders of Justin Gash for this one.)

- Element: An object that is a member of a set.
  - Derek Jeter
  - π
  - ☺

SET OPERATIONS

- Union: $A \cup B =$ the set of elements in A or B
- Intersection: $A \cap B =$ the set of elements in A and B
- Example:
  - $A =$ even numbers between 1 and 10 $= \{2,4,6,8\}$
  - $B =$ prime numbers between 1 and 10 $= \{2,3,5,7\}$
  - $A \cup B = ?$
  - $A \cap B = ?$

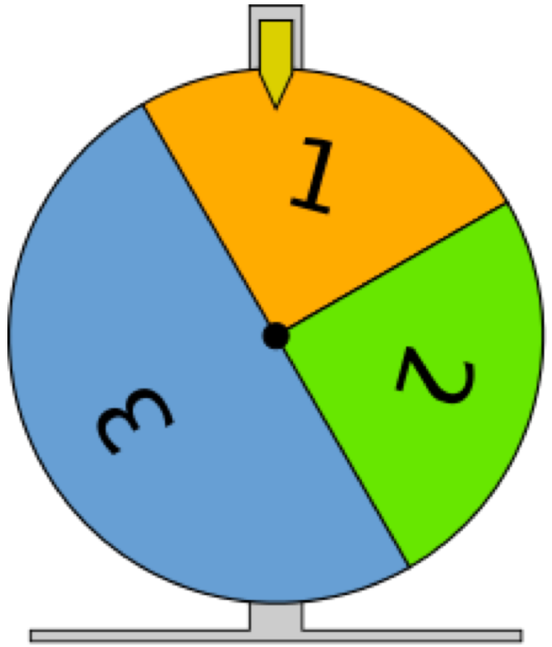
PROBABILITY – PRACTICE

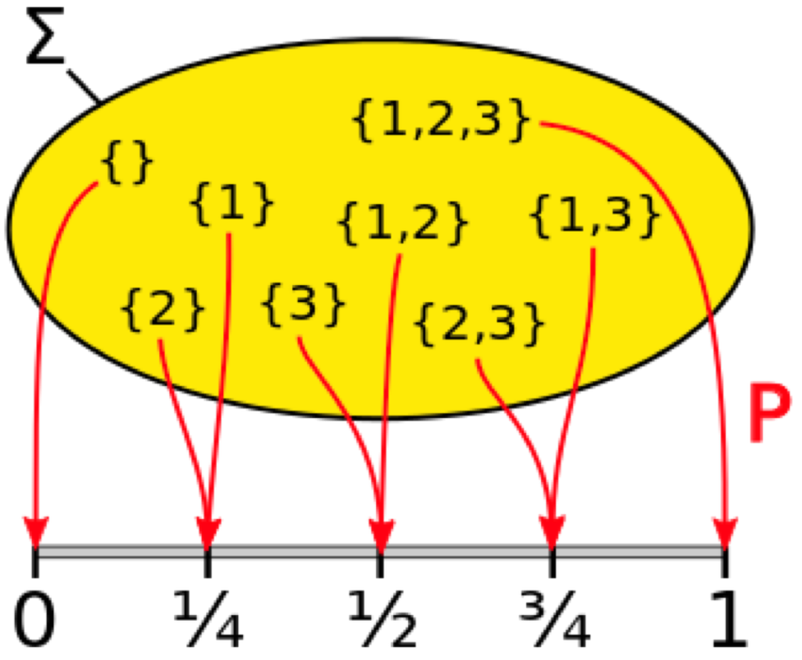
- A = {a U.S. birth results in twin females}
- B = {a U.S. birth results in identical twins}
- C = {a U.S. birth results in twins}

- In words, what does $P(A \cap C)$ mean?

- In words, what does $P(A \cap B \cap C)$ mean?

PROBABILITY BASICS





PROBABILITY RULES

- $P(\emptyset) = 0$
  - Note: $\emptyset$ indicates the "empty set," or the event containing zero outcomes from the experiment.

PROBABILITY RULES

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  - Venn diagrams can help to illustrate this – but remember that Venn diagrams are not proofs!
  - If A and B are disjoint, then $P(A \cap B) = 0 \Rightarrow P(A \cup B) = P(A) + P(B)$.

PROBABILITY RULES

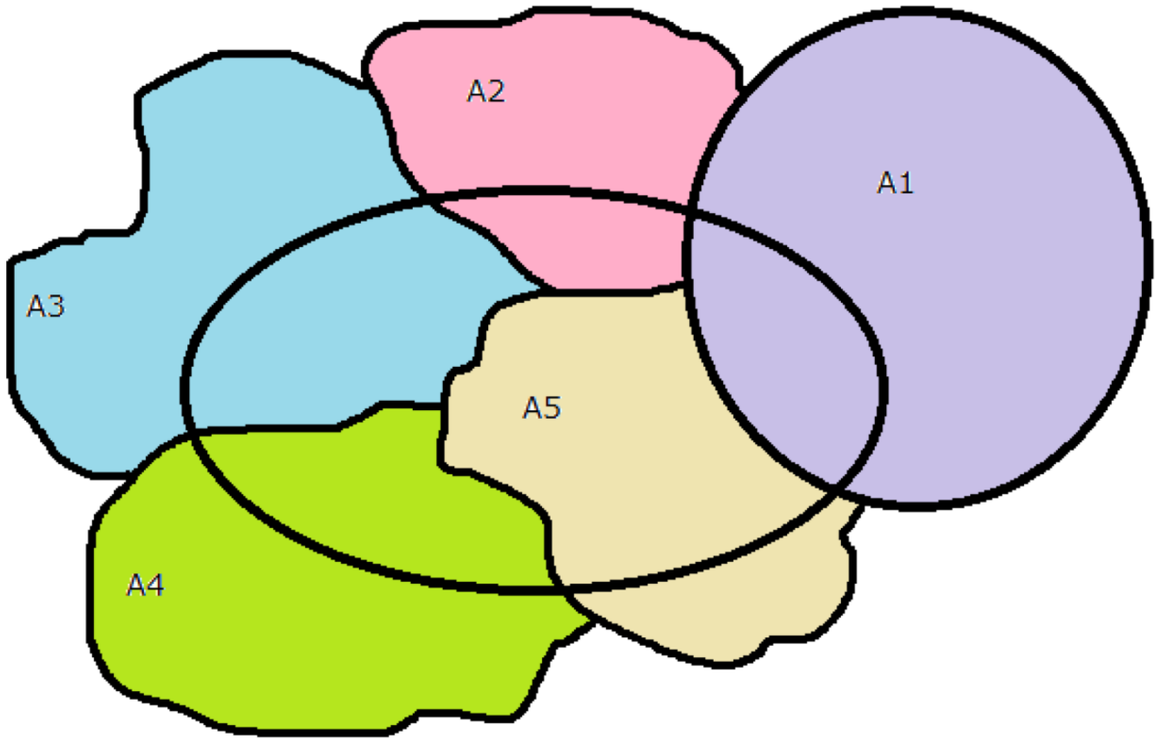
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
  - Note: $A|B$ means "A given B" or "A conditional on the fact that B happens."

PROBABILITY RULES

- $P(A \cap B) = P(A|B)P(B)$
  - We took the last rule, multiplied both sides of $P(B)$, and voila!
  - We can rearrange these, as well! $P(B \cap A) = P(B|A)P(A)$
  - $P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$

PROBABILITY RULES

- $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$
  - "Law of Total Probability"



PROBABILITY RULES – SUMMARY

- $P(\emptyset) = 0$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $P(A \cap B) = P(A|B)P(B)$
  - $P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$
- $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$

BAYES' THEOREM

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

PROBABILITY RULES – PRACTICE

- A = {a U.S. birth results in twin females}
- B = {a U.S. birth results in identical twins}
- C = {a U.S. birth results in twins}

- A twin birth occurs approximately 1 in every 90 births.
- Roughly $\frac{1}{3}$ of all human twins are identical and $\frac{2}{3}$ are fraternal.
- Identical twins are necessarily the same sex and are male with probability 50%.
- Among fraternal twins, $\frac{1}{4}$ are both female, $\frac{1}{4}$ are both male.
- Find the values of $P(A \cap C)$ and $P(A \cap B \cap C)$.
  - Note any assumptions that you make along the way.

WHAT IS $P(A)$?

- We've talked a lot about probabilities of certain events, but what does this actually mean?

- There are two broad classes of probabilistic interpretations.

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- In the long run, how many times will A occur relative to how many times we conduct our experiment?

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- This is called the **frequentist** interpretation of probability.

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- This is called the **Bayesian** interpretation of probability.

WHEN BY HAND IS TOUGH…

- Oftentimes, we won't evaluate probabilities by hand.
  - It's still very important to understand the ideas behind probability – as we move forward, it's critical to:
    - a) know probability's relationship with statistics and machine learning.
    - b) identify potentially bad assumptions.

- However, we can use simulations to give us a good approximation of the true probability of some event.